



# **Advanced Works Like Presentation**

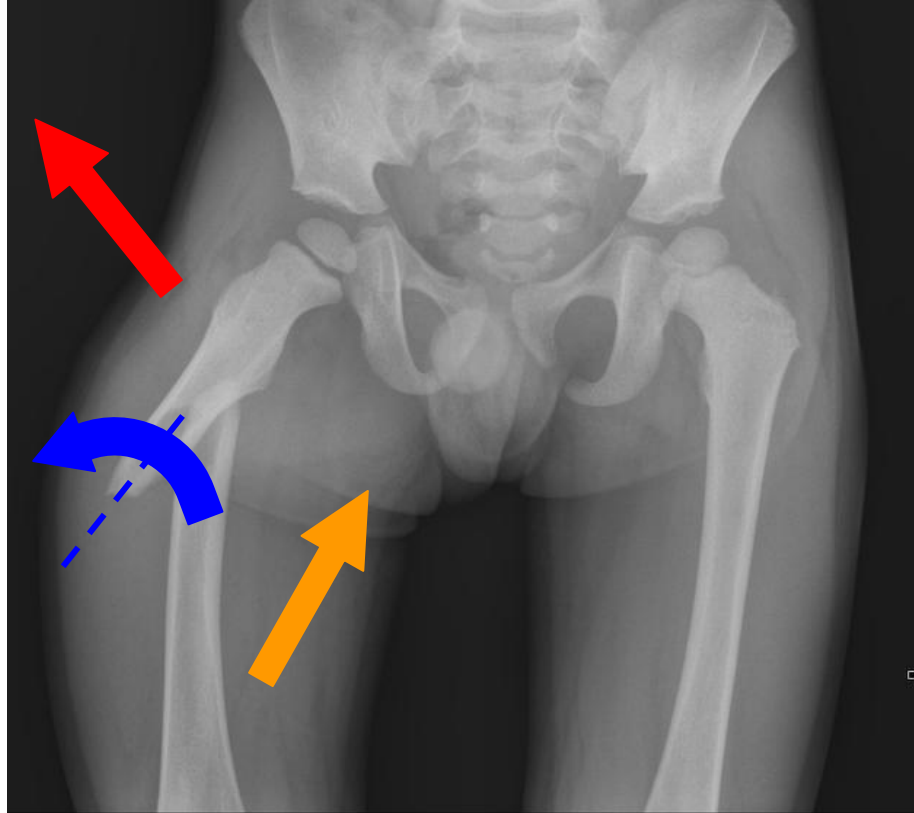
## **Ortho Team**

Bianca, Braxton, Jonathan, Li

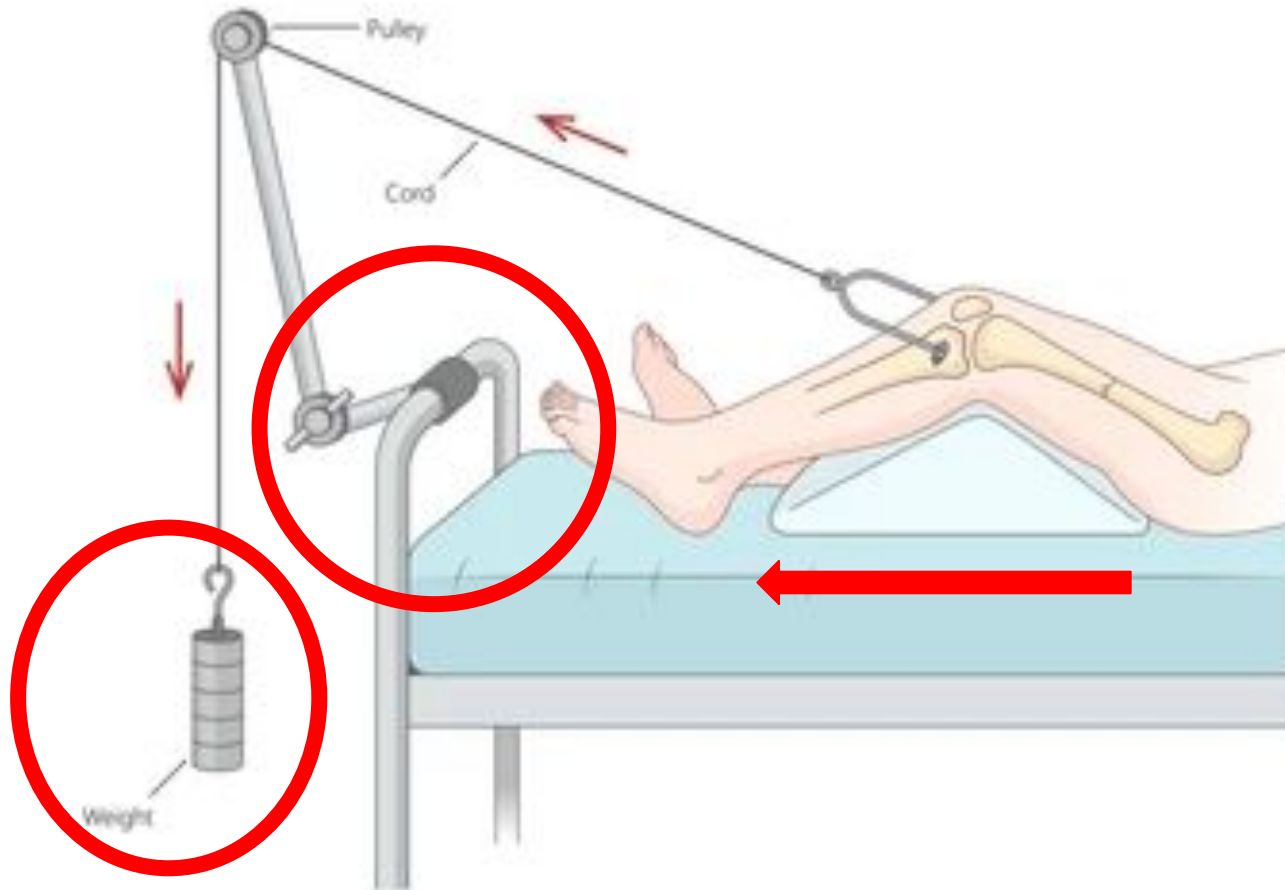
# Background



# Problem Recap



# Skeletal Traction





# Need Statement

Orthopedic surgeons need a way to apply skeletal traction to adult patients with femoral shaft fractures that maintains correct femur length throughout treatment despite changes in patient position to prevent femur displacement that can cause further damage.

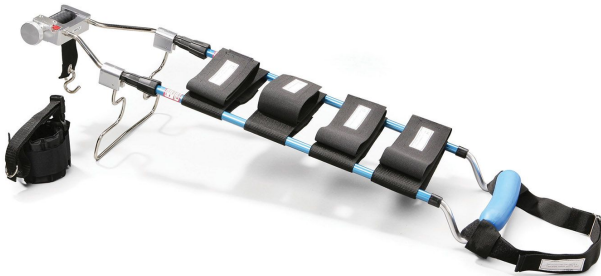
# Current Technology Limitations



Expensive!

Either portable OR  
sufficient traction

No indication of how  
much force is applied



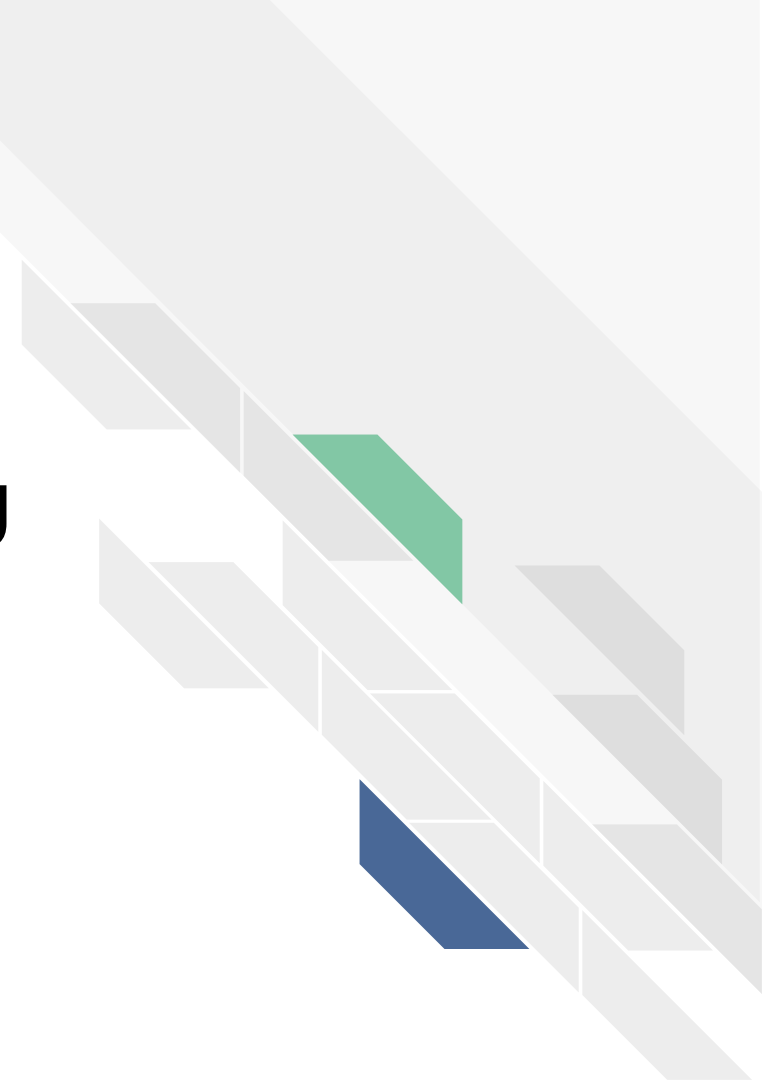


# Value Proposition

Our portable skeletal traction device allows orthopedic surgeons to apply and monitor traction, maintaining correct femur length even **during patient repositioning**.

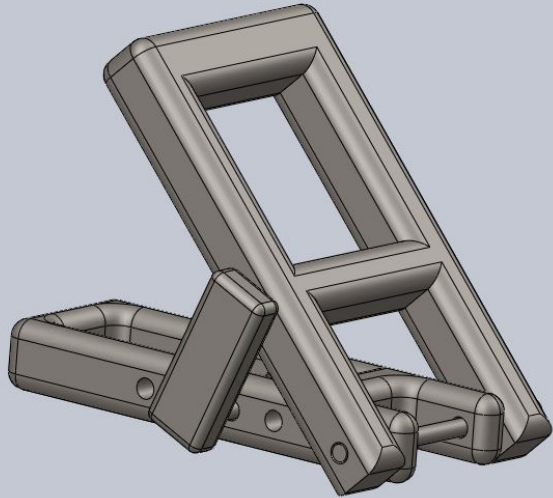
Unlike current methods, the device is **compatible with most standard hospital beds**, and continued **traction throughout scans** enables more accurate treatment plans for better outcomes.

# **CAD Iterative Modeling and FEA Simulations**

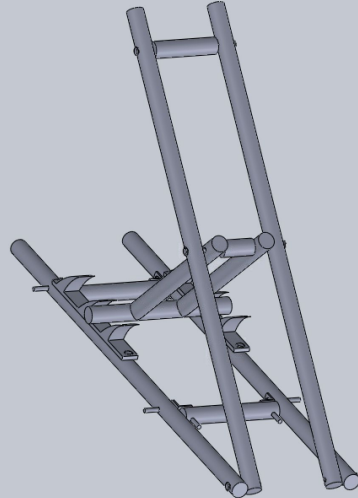


# Sprint Style Rapid Prototyping Iterations

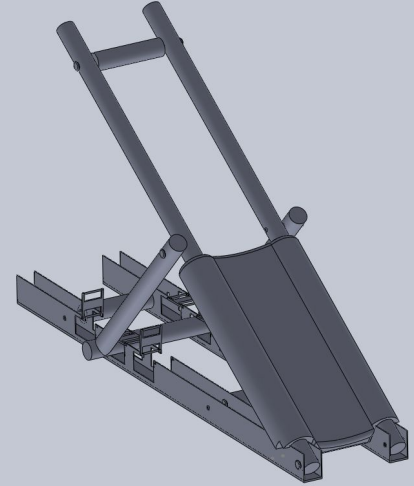
**Iteration 1**



**Iteration 2**



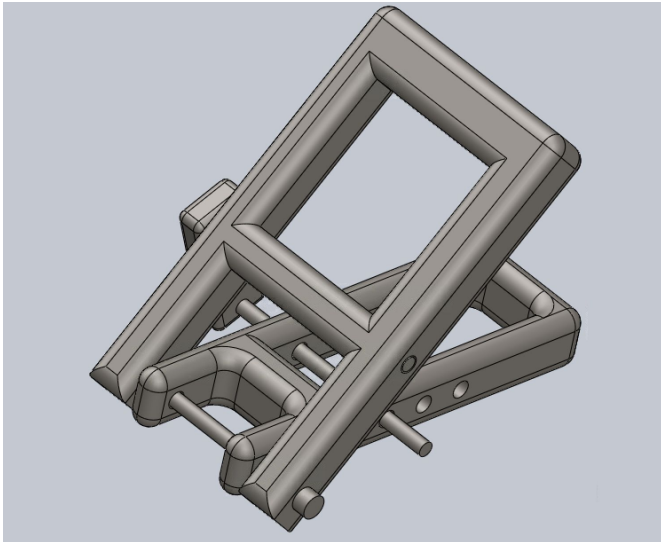
**Iteration 3**



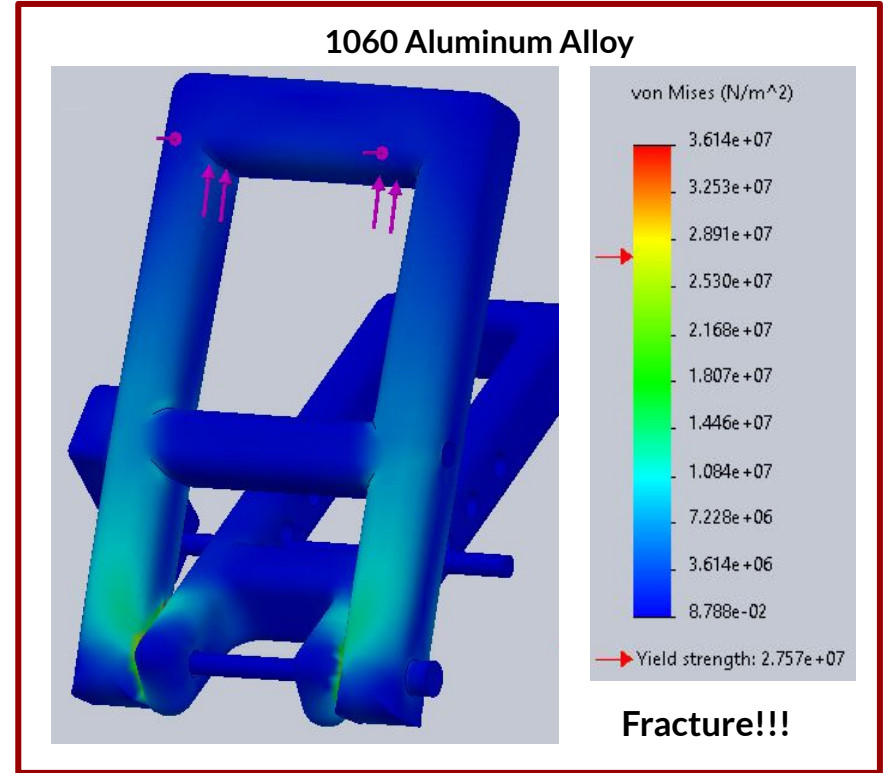
# Iteration 1: Early Mechanical Performance & Design

## Design Gaps

1. Geometry → Fracture
2. Angle Adjustment → Finicky
3. Manufacturing Difficulty



Poor assembly tolerance due to distant hole alignment

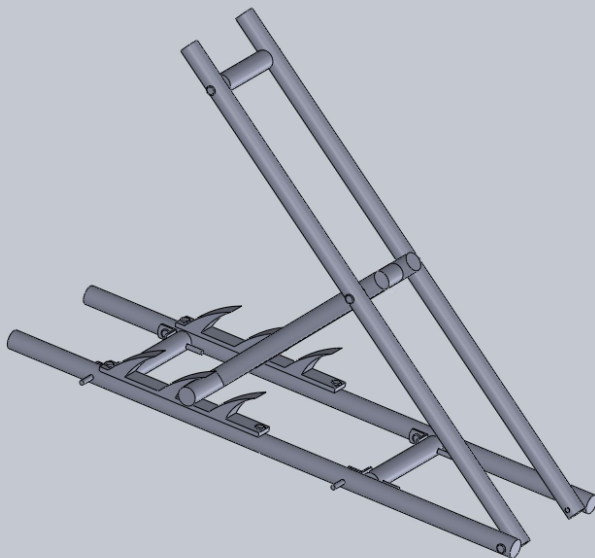


Poor load distribution → high stress at lower joints ( $3.6e7 \text{ N/m}^2$ )

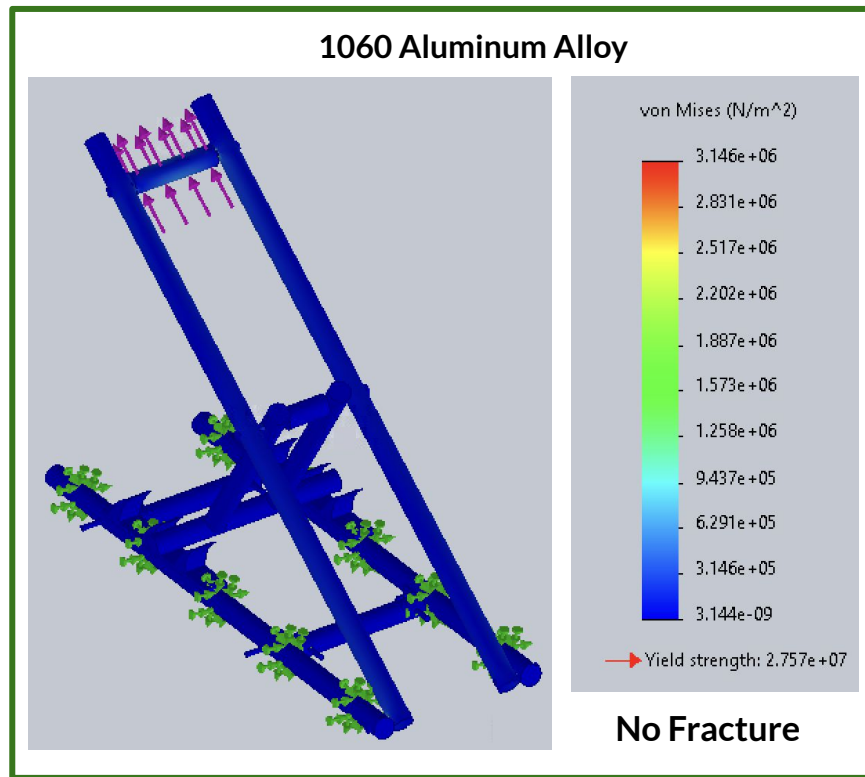
# Iteration 1 → 2: Mechanical Performance & Design

## Design Improvements

1. Geometry → No Fracture
2. Improved Angle Adjustment
3. Easier Manufacturing



Easier alignment for angle adjustment

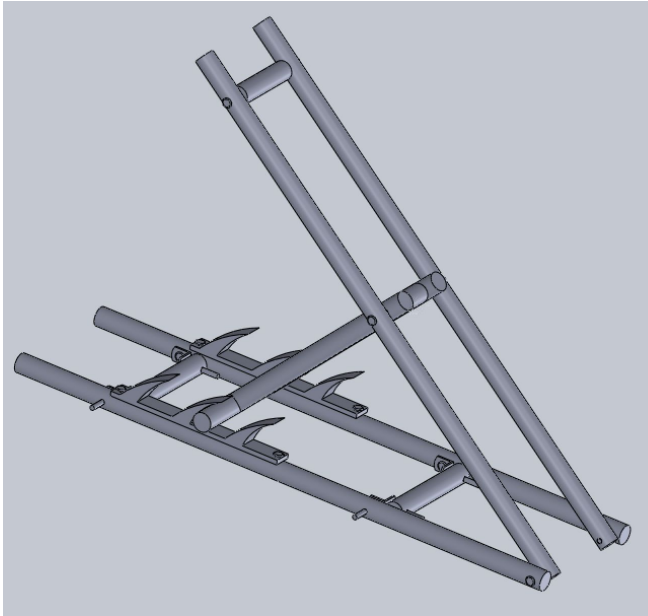


10x Better load distribution → lower stress at joints ( $3.1\text{e}6 \text{ N/m}^2$ )

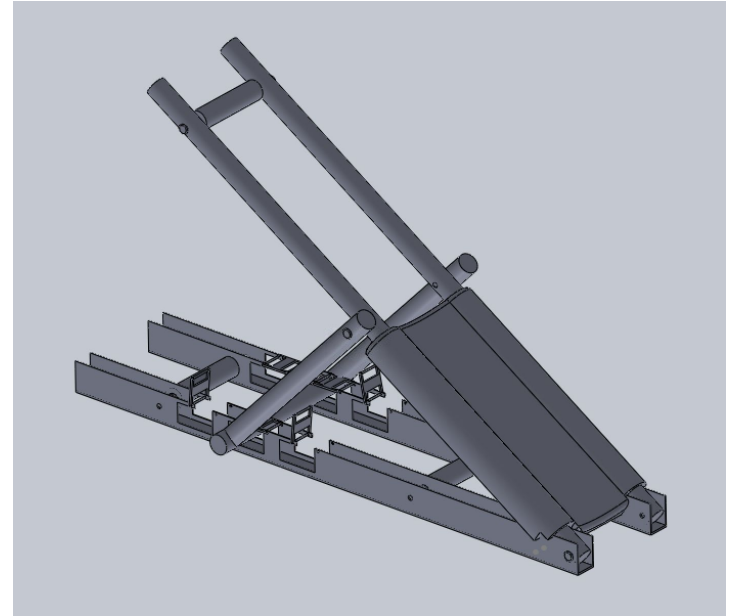
# Iteration 2 → 3: Advancing Clinical Usability

## Design Goals

1. Patient Comfort
2. Medical Staff Usability



Iteration 2: Design for mechanical performance

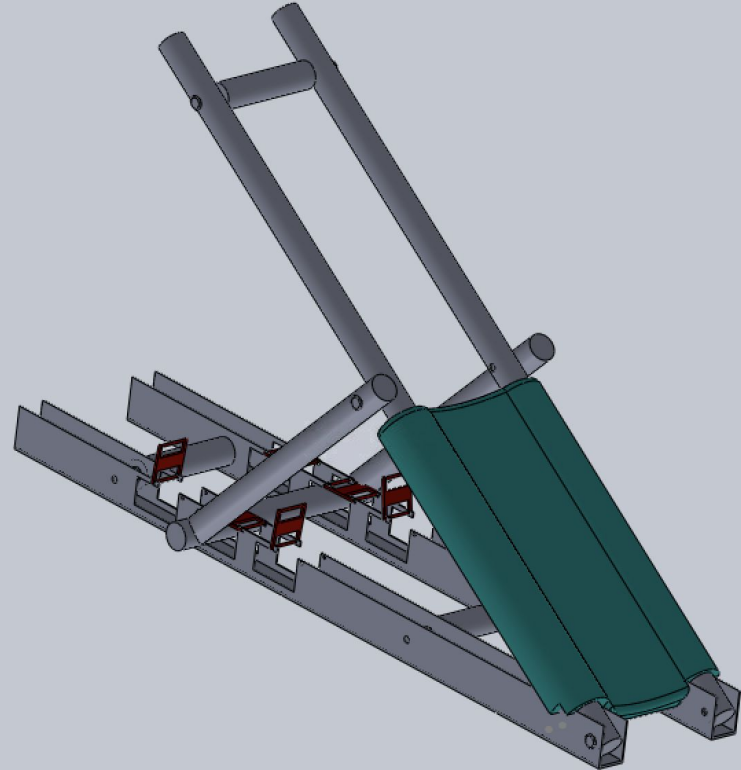
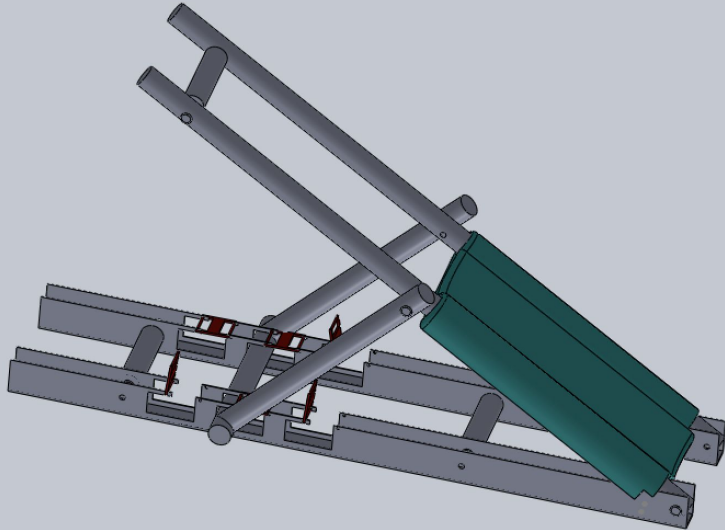


Iteration 3: Design for clinical application

# Iteration 3: Designing for Clinical Usability

## Design Implementations

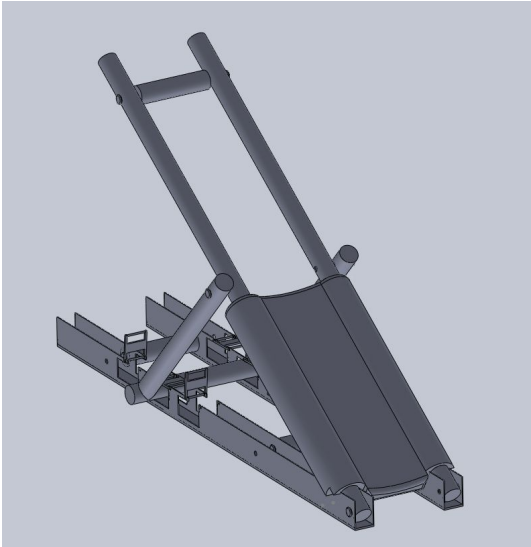
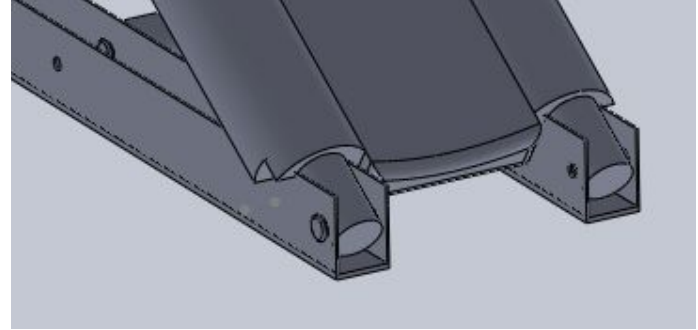
1. Leg Rest
2. Angle Adjustment Clips



# Iteration 3: Clinical Design Gaps

## Design Gaps

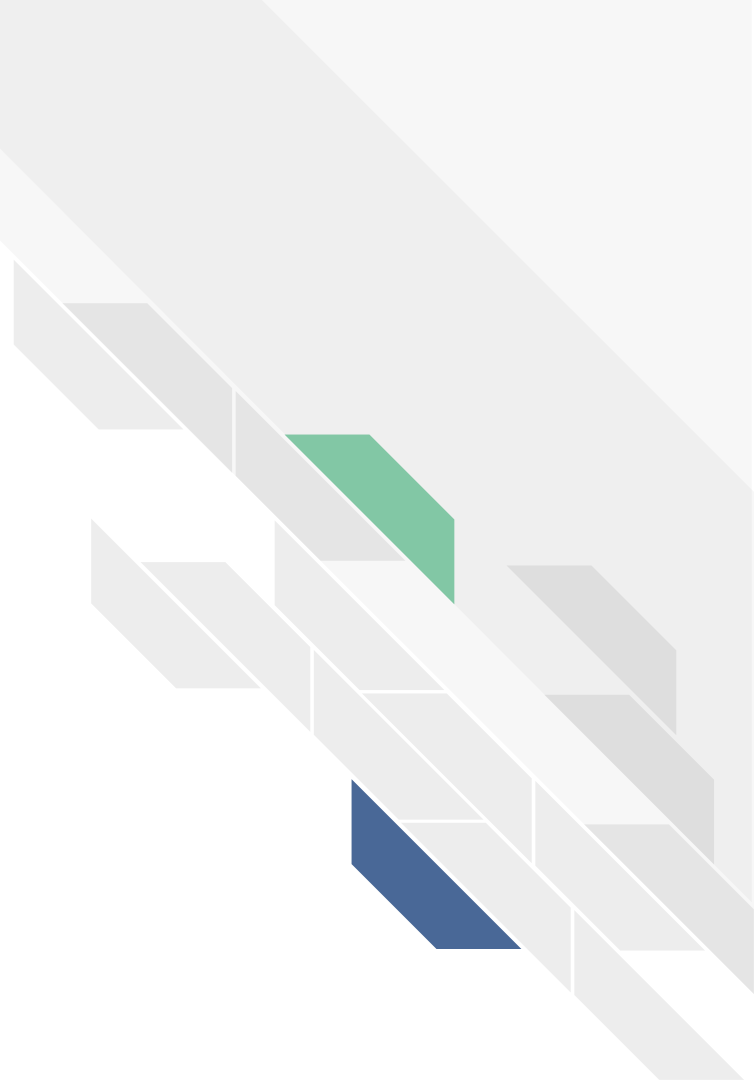
1. Padding at lower joints
2. Clamps to fasten fixture to bed (stabilization)



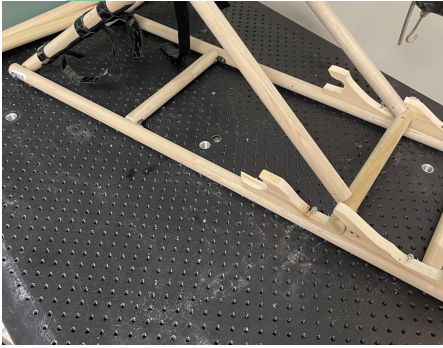
Stability with standard bed



# Physical Prototype



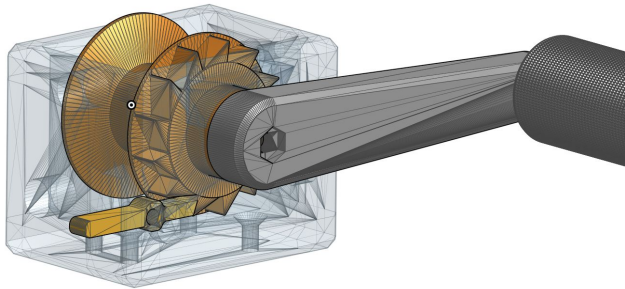
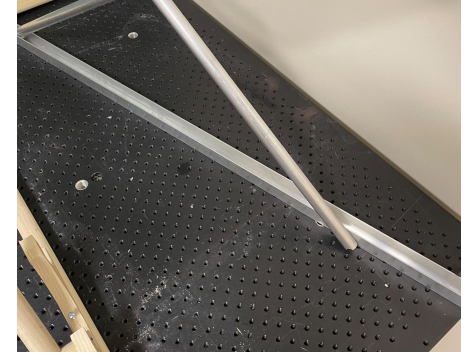
# Initial vs Advanced Works Like Prototype



↑ Collapsibility  
↑ Stability



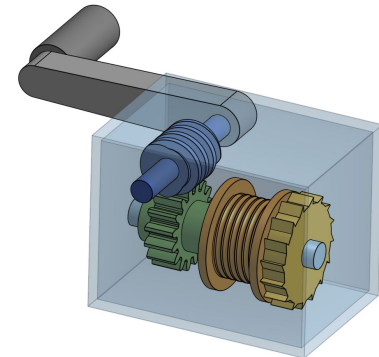
Base Frame Rod → U-Channel



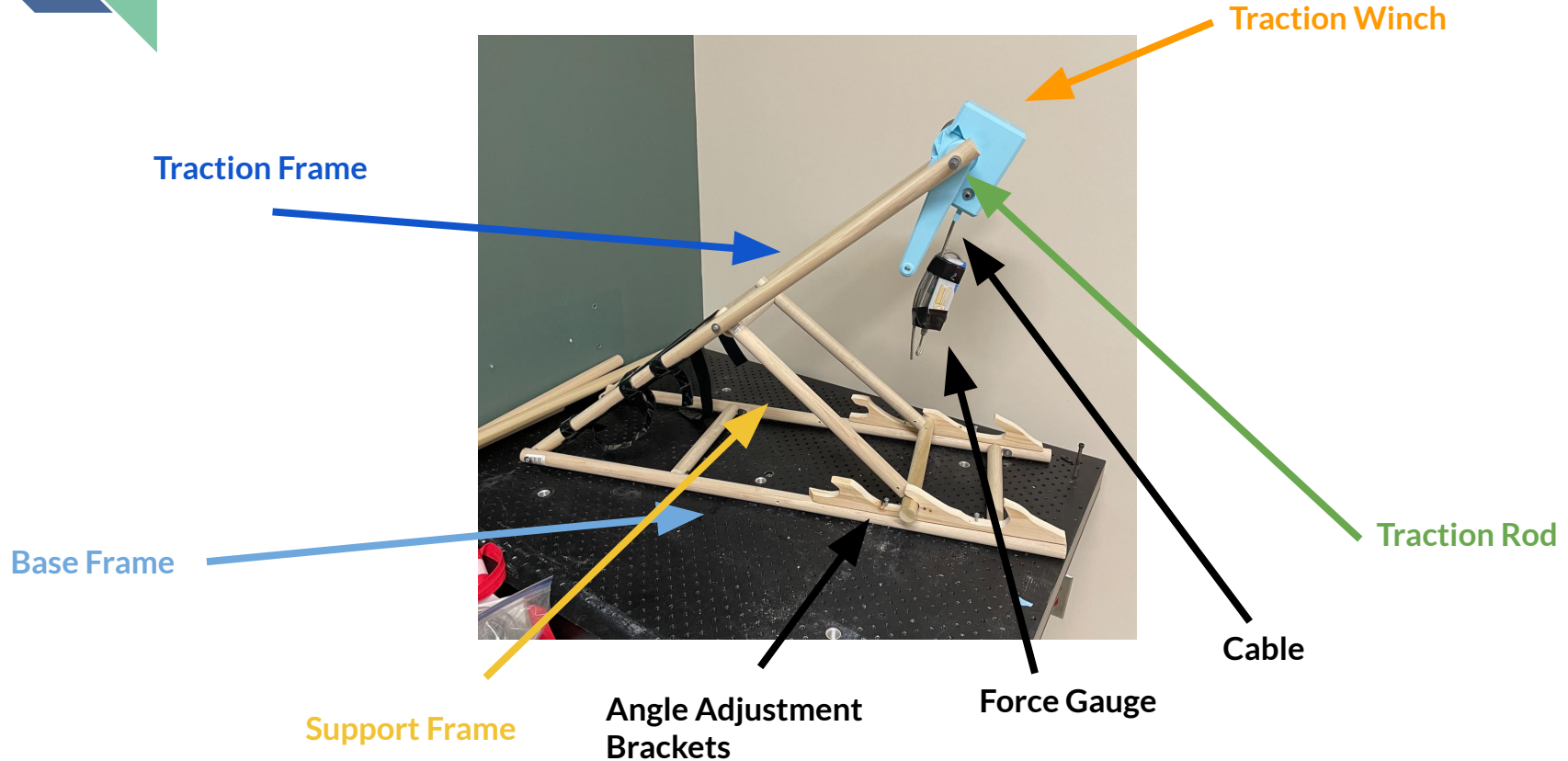
↑ Usability  
↓ Risk of injury to patient and user



Winch Design



# Advanced Works Like Prototype



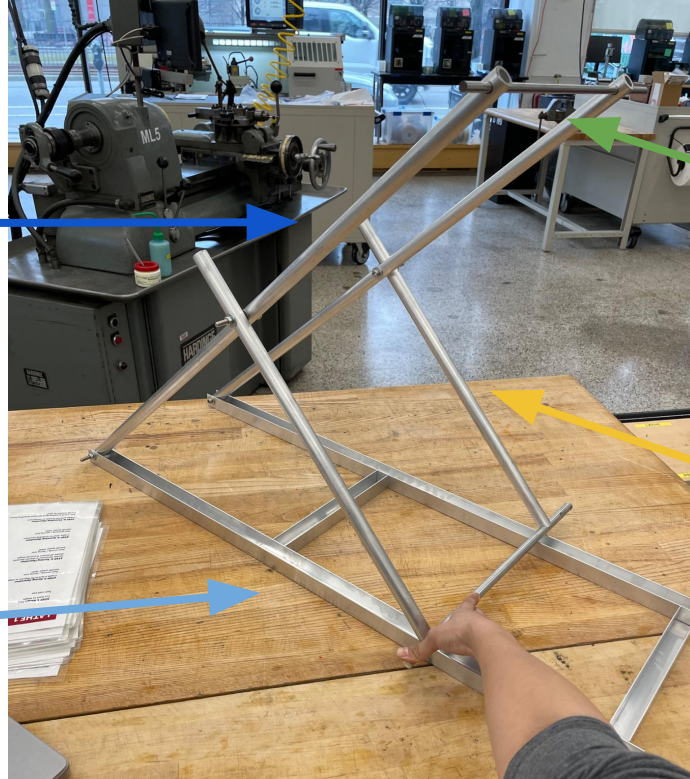
# Advanced Works Like / Is Like

Traction Frame

Traction Rod

Support Frame

U-Channel Base Frame





# Proposed Solution

|    | User Need                                                                                |
|----|------------------------------------------------------------------------------------------|
| 1. | Device shall apply traction to femur fracture per current treatment requirements         |
| 2. | The device shall maintain the set length and position of femur throughout treatment      |
| 3. | User must be able to know forces applied                                                 |
| 4. | Device shall be radiolucent for X-rays and CT scans                                      |
| 5. | The device must be compatible with use of most hospital bed, including X-ray and CT beds |



Can it apply a pulling force on bone pin? **YES!**



Can maintain traction? **YES!**



Can the amount of applied force be measured? **YES!**



Can it be used in scans? **YES! (material)**



Can it be used with different beds? **YES!**

# Initial Bill of Materials (BOM) - Frame

**TOTAL = ~\$220.49**

## **Traction Frame** and **Support (Vertical)**

6061 Aluminum Round Tube ( $\frac{3}{4}$ " OD,  $\frac{1}{8}$ " T)  
7ft 8in + 3ft  $\Rightarrow$  ~\$78.35

Force Gauge (not shown)  
~\$100

## **Traction Bar**

Stainless Steel Rod (0.3" OD)  
1 ft  $\Rightarrow$  \$7.84

## **Support (Horizontal)**

Aluminum Round Rod (0.5" OD)  
1 ft  $\Rightarrow$  \$2.68

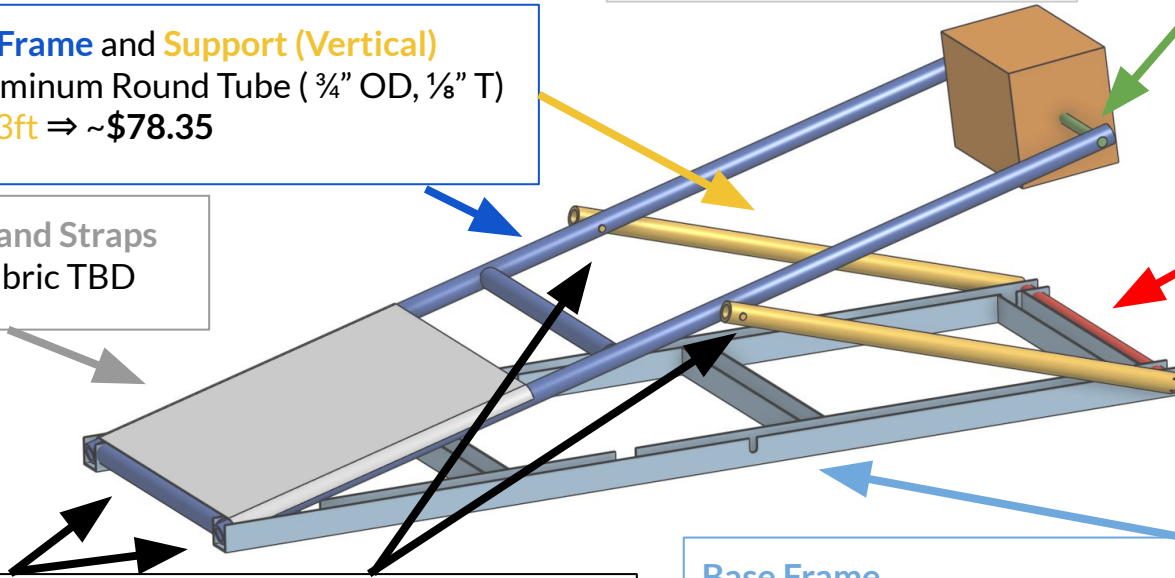
**Leg Cradle and Straps**  
Plastic or fabric TBD

## **Hinges/Pivot Points**

Pin and Bushing with Loctite 638  
(More research required for exact materials/price)

## **Base Frame**

6061 Aluminum U-channel (1"x1" Outer dimensions,  $\frac{1}{16}$ " T)  
3ft 2in  $\Rightarrow$  ~\$38.62



# Initial Bill of Materials (BOM) - Winch

**TOTAL = ~\$47.18**

## Worm

1045 Carbon Steel Rod (1  $\frac{3}{8}$ " OD)  
1.42 in @ \$22.29/1ft = **\$2.67**  
+  
 $\frac{5}{8}$ " OD  
2 in @ \$9.98/3ft = **\$0.55**

## Worm Gear

6061 Aluminum Sheet (4" W, 1" T)  
12.25 sq in @ \$0.76/sq in = **\$9.31**

## Housing

6061 Aluminum Sheet (0.16" T)  
121 sq in @ ~\$0.24/sq in = **\$20.47-\$28.66**

## Crank

6061 Aluminum Shaft, handle  
material TBD

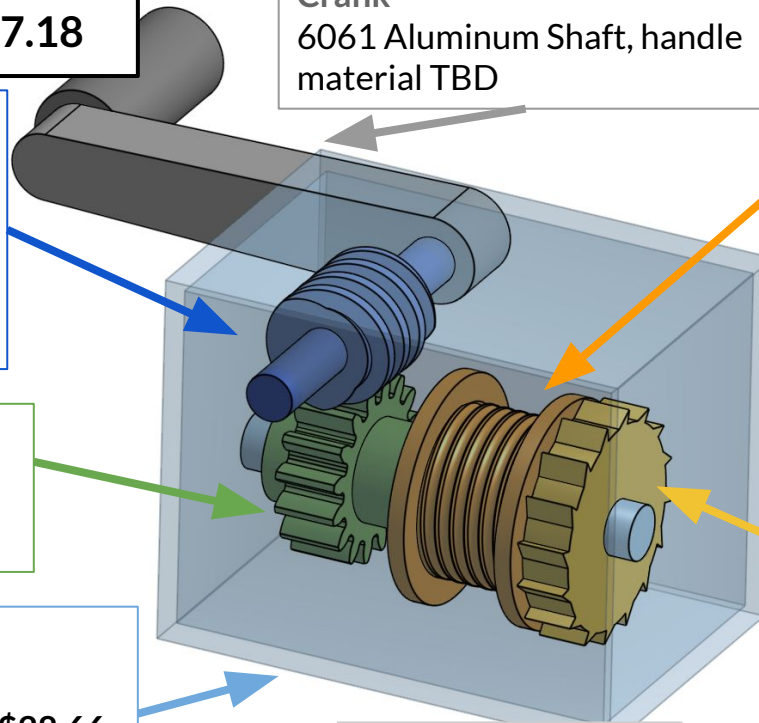
## Drum

6061 Aluminum Rod (2" OD)  
1 in @ \$25.49/6 in = **\$4.29**  
+  
6061 Aluminum Sheet (4"W,  $\frac{1}{4}$ " T)  
5.3 sq in x 2 @ \$0.24/sq in = **\$2.54**

## Ratchet and Pawl

1045 Carbon Steel Sheet  
(12"x12"x $\frac{3}{8}$ " T)  
12.25 sq in @ \$0.60/sq in  
= **\$7.35**

**Cable (not shown)**  
Material TBD





# Cost Overview

**Materials: \$267 (frame = \$220, winch = \$47)**

+ Excluded materials (leg cradle/straps, winch crank, cable)

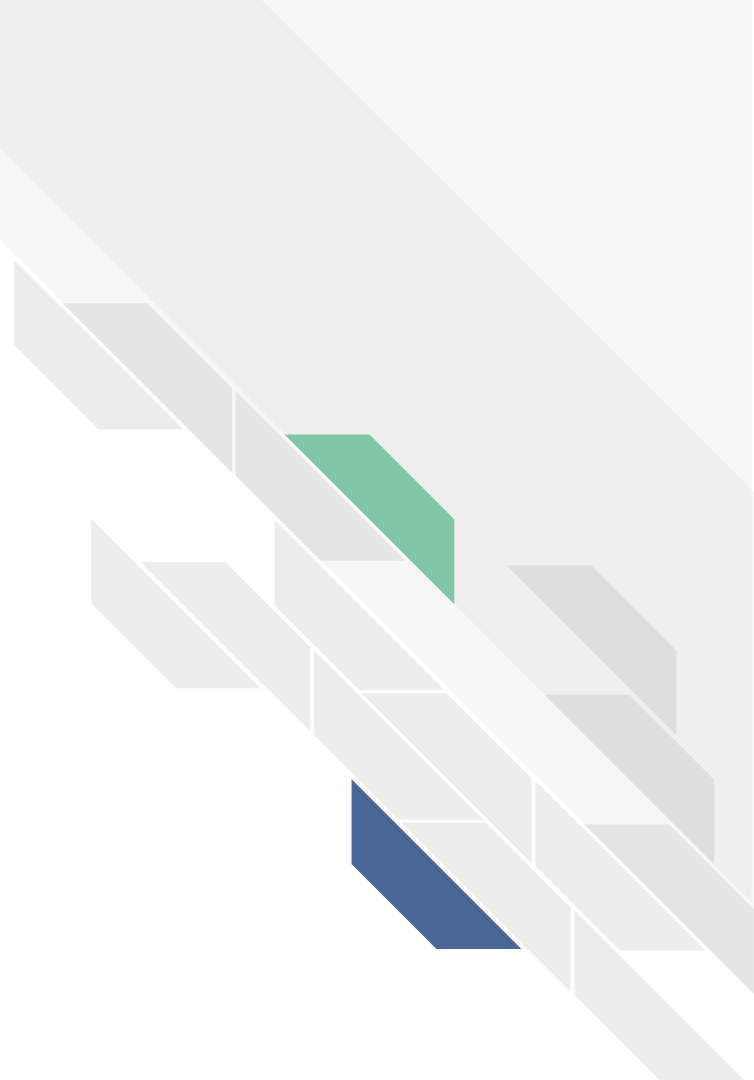
+ Machining and assembly time

However,

- raw material mass production costs are lower

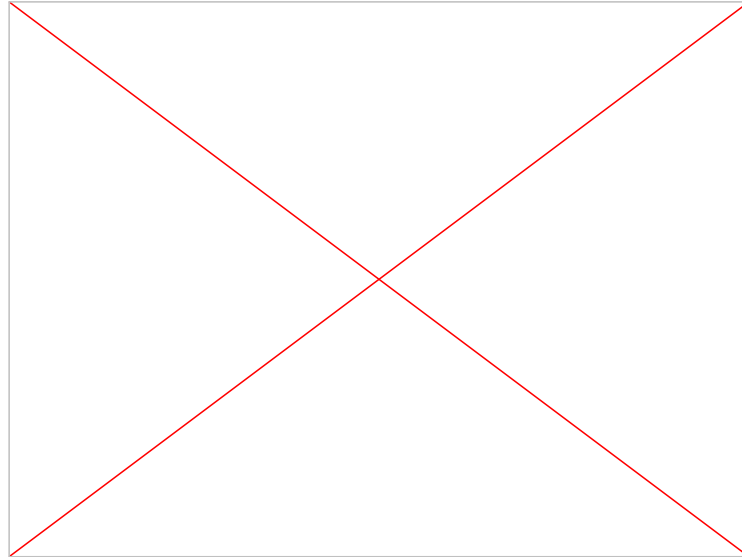
- winch parts may be bought off the shelf

# Verification Testing



# Initial Verification - Winch

- Testing prototype capability to adjust traction force through turning the winch and achieve 30 lbs of traction



# Initial Verification - Winch

- Takeaways:
  - Functions as intended
  - Winch handle repositioning
  - More obvious user feedback when turning the winch



# Initial Verification - Applied Traction

- Determining if traction force will fluctuate/creep over time outside of  $\pm 1$  lb range for 30 min
- Takeaways:
  - Minimal fluctuations within desired range
  - Stabilize patient leg
  - Simulate experimental noise in future testing environment



# Validation Plan - Frame Portability

- Confirm clinical staff can carry frame and maneuver around necessary areas
- Test whether clinical staff can transfer patient stand-in traction from bed to scanning table



# Validation Plan - Traction Workflow

- Assess time needed for staff to apply or take off traction to patient stand-in compared to SOC
- Ensure staff can turn winch and accurately set to target weight +/- 2 lbs



# Validation Plan - X-Ray Compatibility

- Test frame compatibility with standard femur fracture scanning workflow, measuring time needed to complete tasks
- Scan a frozen chicken in and out of traction to evaluate signal obstruction



# **Failure Modes and Effects Analysis**





## Failure Modes: Top 3

1. Frame breaks due to large amount of load
2. Winch unlocks when device is in use
3. Damage to the nerve



**top 3**

# Failure Modes:Top 3

1. Frame breaks due to large amount of load

## Effects:

- Interruption of treatment, need of alternate device

## Mitigation Plan:

- Label correctly the limits of the device
- Design for the device to fit the 90% of the adults in the US population
- Specify in the IFU that the limit of load is 10% of the patient's body weight or 40lbs



# Failure Modes:Top 3

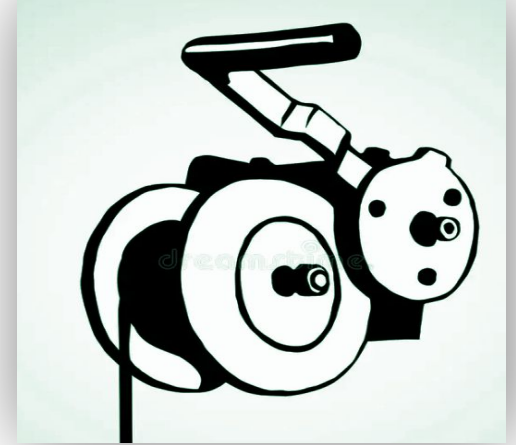
## 2. Winch unlocks when device is in use

### Effects:

- Traction is not applied, patient is injured severely and fracture complicates

### Mitigation Plan:

- Specify in the IFU locking mechanisms instructions
- Add a sensor light that indicated device has locked correctly
- Create a double locking mechanism in case one fails



# Failure Modes: Top 3

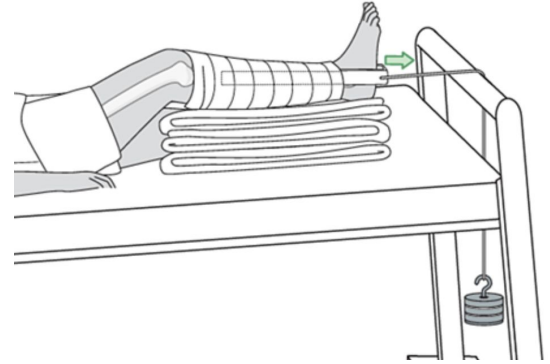
## 3. Damage to the Pudendal nerve

### Effects:

- Nerve gets pressured for a large amount of time and constant force which can result in neuropathy, chronic pelvic pain and numbness

### Mitigation Plan:

- Add padding to the frame to minimize effects



# Killer Experiment



- Traction Workflow
- Methods: Apply traction to patient leg, transfer frame + patient to scanning table, take scans + remove traction
- Metrics: NASA Task Load and imaging intensity
- Control group: standard skeletal traction



# Challenges and Next Steps

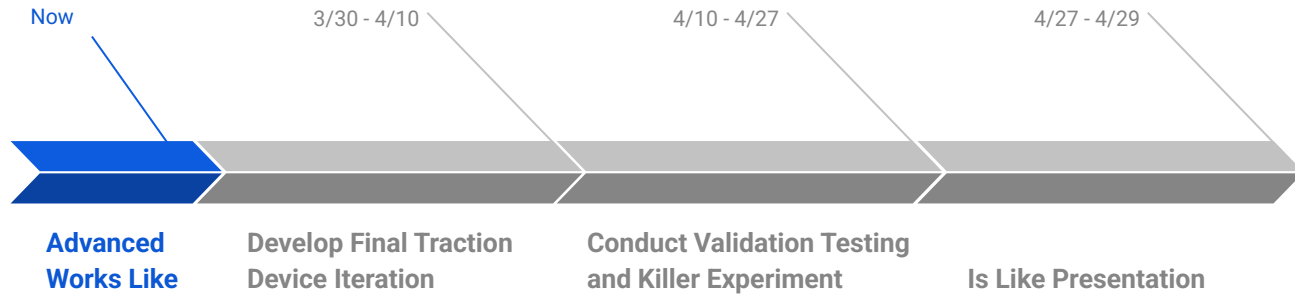
## Challenges

- Balancing patient comfort with device stability and core functionality
- Keeping device affordable for consumers
- Design for Manufacturing

## Next Steps

- Develop final device iteration
- Match Solidworks dimensions to physical device for precise FEA analysis
- Conduct validation testing
- Evaluate killer experiment
- Finalize development of business plan

# Timeline



**Questions?**

